1. Ratha has two balls less than Kanha. If Kanha has ball ∇ , how many balls does Ratha have?

   A) $2\times\nabla$ **B)** $\nabla-2$ **C)** $2-\nabla$ **D)** $2\div\nabla$

2. $\frac{7}{8}$ of the whole pizza in Sok refrigerator. He eats $\frac{3}{8}$ of the whole pizza for lunch. What is the fraction of a whole pizza is left after Sok eating pizza for lunch?

   A) $\frac{4}{8}$ **B)** $\frac{3}{8}$ **C)** $\frac{5}{8}$ **D)** $\frac{10}{8}$

3. Which of the following number has the same value of 16×10^6 ?

   A) 0.16 **B)** 160 **C)** 0.000016 **D)** 16000000

4. Which x value makes the following inequality true?

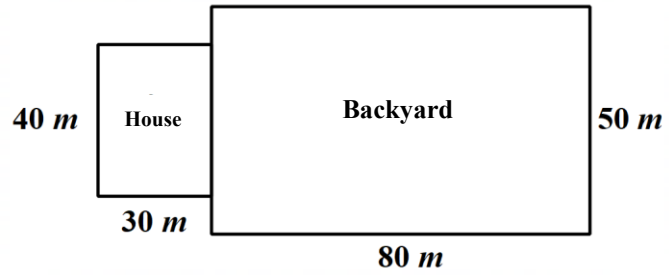
$$3x-4<11$$

   A) $\frac{13}{2}$ **B)** 6 **C)** 3 **D)** 5

5. David learns to cook for $ 120.00 a month. Which equation can be used to determine the total amount of dollars "d" that David spends on a monthly lesson "m"?

   A) $m=120+d$ **B)** $d=120\times m$ **C)** $m=120\times d$ **D)** $d=120+m$

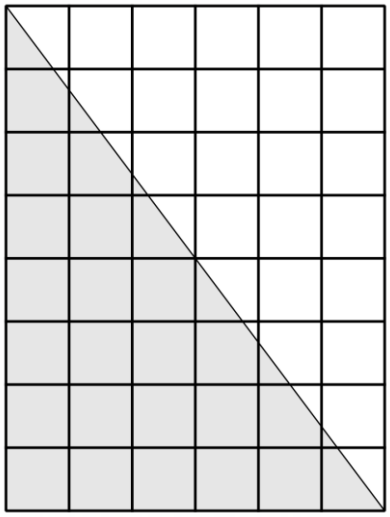
6. The diagram below shows the dimensions of a rectangular house with a backyard of a rectangular house.



   What is the total area of the house and the backyard in square meter?

   A) 5,200 **B)** 4,000 **C)** 200 **D)** 4,200

7. The grid shown below is rectangular.



What is the area of a square unit of the shaded section of a rectangle?

A) 14 **B) 36** **C) 48** **D) 24**

8. Building a fence around a square plot of land with a side length of 80 m. The finished fence is 218 m long. How many meters of fence is left to be completed?

A) 100 m **B) 90 m** **C) 102 m** **D) 150 m**

9. Which number will be in the tens place after 61,324 divided by 4?

A) 1 **B) 3** **C) 4** **D) 2**

10. Veasna has $ 0.25 in his phone. He topped up $ 5. He calls his wife for $ 0.750, calls his children for $ 1.05 and sends text messages to his friends for $ 0.50. How many dollars is left in his phone?

A) 2.95 $ **B) 3.15 $** **C) 2.15 $** **D) 1.98 $**

11. Which two expressions are equivalent?

A) $12x^2+5x+10$ and $17x^2+10$ **B)** $x+x+x$ and x^3

C) $14x-10-2x$ and $16x+10$ **D)** $12x+16x$ and $4(3x+4x)$

12. What is the value of $\frac{5}{3}-\frac{1}{2}\div\frac{3}{4}$?

A) $\frac{2}{3}$ **B)** 2 **C)** 1 **D)** $\frac{3}{2}$

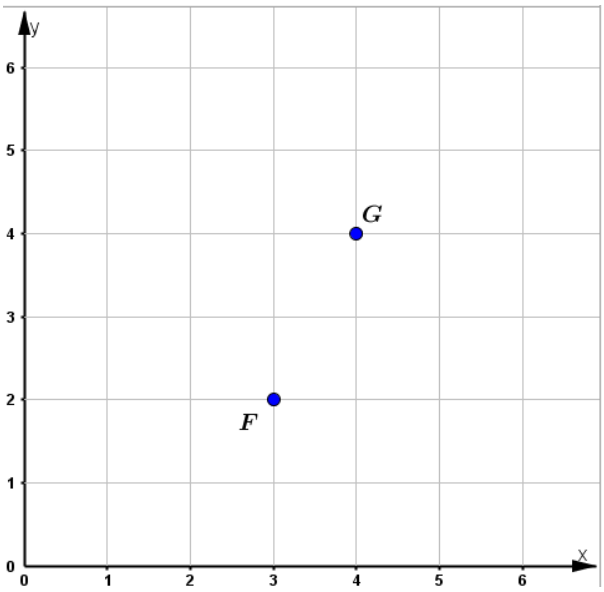
13. A family of two adults and three children went to the movies. Tickets are \$ 12 for adults and \$ 6 for children. Which expression represents the total ticket price?

A) $2+3+12+6$ **B)** $(6\cdot12)+(3\cdot2)$ **C)** $(3\cdot12)+(2\cdot6)$ **D)** $(2\cdot12)+(3\cdot6)$

14. Which expression represents the prime factor?

A) $2\cdot2\cdot2\cdot2\cdot3\cdot11$ **B)** $2\cdot9\cdot11$ **C)** $2.5\cdot7\cdot3$ **D)** $1\cdot11\cdot13$

15. Points F and G are shown on the coordinate plane below.



Point G and point H are the same distance from point F. Which coordinate can be the position of point H?

A) $(2,5)$ **B)** $(5,1)$ **C)** $(4,2)$ **D)** $(1,2)$

16. In the figure below, what are the two numbers α , β ?

$$28 ; 23 ; 18 ; 13 ; \alpha ; \beta$$

A) $6 ; 1$ **B)** $9 ; 5$ **C)** $10 ; 7$ **D)** $8 ; 3$

17. One seller sold $ 620,000 last year, which is 70% of the sales she has this year. Which equation can be used to determine the total number of sales in dollars for this year?

A) $\frac{620,000}{100} = \frac{x}{70}$ **B)** $\frac{70}{620,000} = \frac{x}{100}$

C) $\frac{620,000}{x} = \frac{70}{100}$ **D)** $\frac{70}{100} = \frac{x}{620,000}$

18. If $315\frac{1}{5} + 401\frac{2}{5} = \varphi\frac{3}{5}$. Thus, the value of φ is equal to:

A) 812 **B)** 716 **C)** 618 **D)** 418

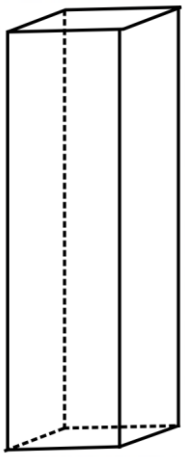
19. Residents of a small town voted on whether to allow developers to build Technology Centers. The number of votes cast for the Technology Center was 4,400. The number of votes against the Technology Center was 17,600. What percentage of voters support the construction of Technology Centers?

A) 20% **B)** 44% **C)** 35% **D)** 25%

20. All of Ratha's workouts consist of 10 minutes of warm-up, 25 minutes of weightlifting and 15 minutes of running on a treadmill. What is the ratio of the number of minutes he lifts weights to the number of minutes of his entire workout?

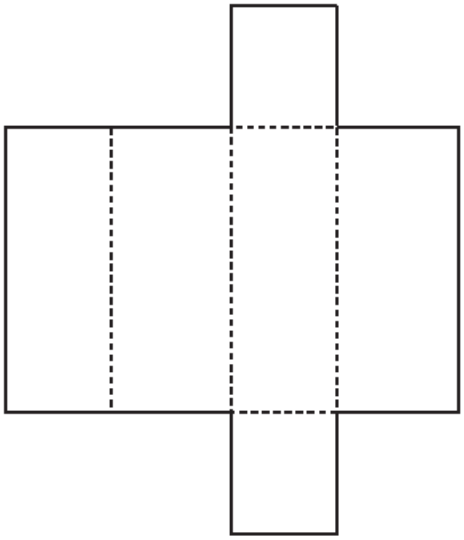
A) 3 : 10 **B)** 3 : 4 **C)** 1 : 2 **D)** 5 : 2

21. The rectangular prism is shown below.

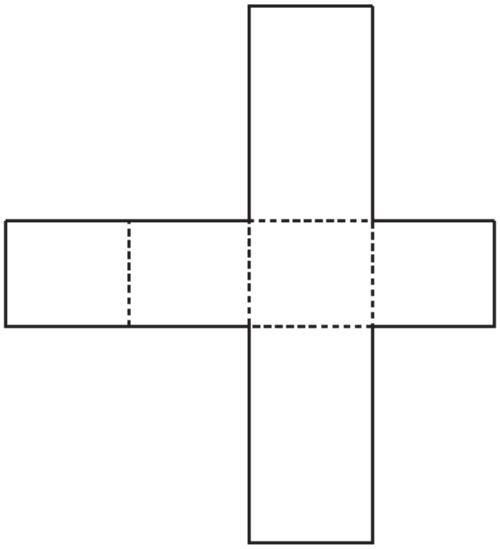


Which figure represents the net of a rectangular prism?

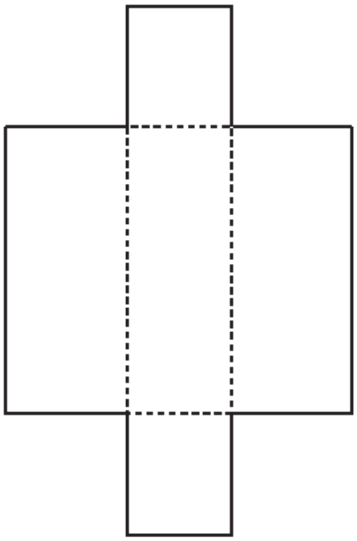
A)



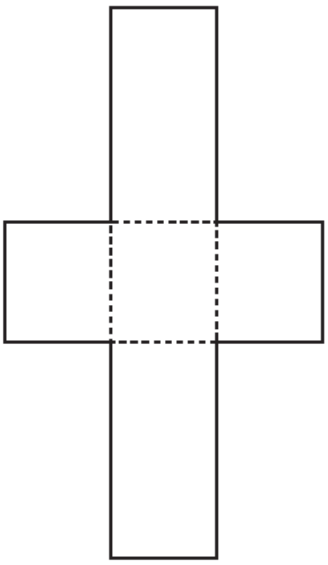
B)



C)



D)



22. Socheata's mother's recipe for making cookies requires 4 kg of sugar and 10 kg of rice flour. Using the same recipe, how many kilograms of sugar does she need for 1 kg of rice flour?

A) $\frac{2}{3}$ kg **B)** $\frac{1}{2}$ kg **C)** $\frac{5}{2}$ kg **D)** $\frac{2}{5}$ kg

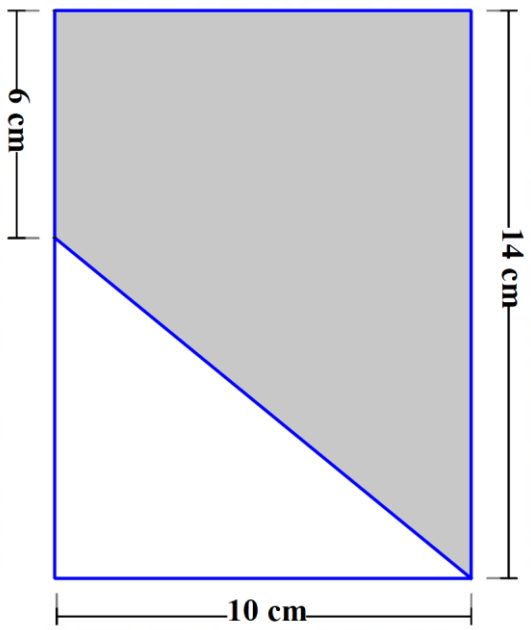
23. What is the greatest common factor of 56 and 92?

A) 8 **B) 4** **C) 2** **D) 6**

24. Dara aims to run a total of 150 km this month. Every day he runs 6 km. Which expression can Dara use to determine how many miles he has left to run after running for a "d" day?

A) $6d$ **B)** $\frac{150}{6d}$ **C)** $150-6d$ **D)** $6d+150$

25. What is the area in square centimeters of the shaded part of the rectangle shown below?



A) 100 **B) 140** **C) 60** **D) 146**

26. In a concert, 40% of the audience are teenagers. If the number of teenagers in the concert is 1460, what is the total audience?

A) 5840 **B) 365** **C) 730** **D) 3650**

27. What is the value of the following expression when $\alpha = 5$ and $\beta = 4$?

$$6\alpha^2 - 5\beta + 8$$

A) 138 **B) 79** **C) 48** **D) 218**

28. Which of the following expression is equal to $18x + 27y$?

A) $2(9x + 17y)$ **B)** $9(2x + 3y)$ **C)** $9x(2 + 7y)$ **D)** $2y(9x + 27)$

29. Sophea works as a cashier at the supermarket, she works 3 hours every night for 12 nights. She earned a total of $ 288. Sophea wants to calculate her hourly rate. How much does Sophea earn per hour?

A) $10 **B) $16** **C) $8** **D) $24**

30. Which of the following equation is correct when $b = 6$?

A) $\frac{12}{b} = 4$ **B)** $4b = 20$ **C)** $3b = 16$ **D)** $b + 8 = 14$